

Counterexample to the c_0 Vector-Measure Range Problem

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Source Problem

The source is Jose Rodriguez, *Open problems in Banach spaces and measure theory*. The arXiv identifier is 1607.07787.

Problem 4.2 appears on PDF page 12 and is attributed there to Anantharaman and Diestel:

Let K be a weakly compact subset of c_0 . Is K contained in the range of a c_0 -valued vector measure?

12

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Problem 4.2 (Anantharaman-Diestel, [4]). *Let K be a weakly compact subset of c_0 . Is K contained in the range of a c_0 -valued vector measure?*

Certain properties of a vector measure are determined by its range. For instance, if ν_1 and ν_2 are vector measures such that $\mathcal{R}(\nu_1) = \mathcal{R}(\nu_2)$, then $|\nu_1|$ is finite if and only if $|\nu_2|$ is finite, [109]. In such case, ν_1 is Bochner differentiable with respect to $|\nu_1|$ if and only if ν_2 is Bochner differentiable with respect to $|\nu_2|$, [110]. An example of Fremlin and Talagrand [61] was used in [111] to show that the previous result is not true in general for the Pettis integral

Counterexample

Let

$$\mathcal{S} = \{\emptyset\} \cup \{F \subseteq \mathbb{N} : 0 < |F| \leq \min F\}.$$

For $F \in \mathcal{S}$, write $1_F \in c_0$ for its characteristic vector. Define

$$K_{\mathcal{S}} = \{1_F : F \in \mathcal{S}\} \subseteq c_0.$$

Theorem 1. *The set $K_{\mathcal{S}}$ is weakly compact in c_0 , but it is not contained in the range of any countably additive c_0 -valued vector measure.*

Idea of the Proof

The Schreier family is compact as a family of finite subsets of $\mathbb{N}$, so $K_{\mathcal{S}}$ is weakly compact in c_0 . Suppose nevertheless that $K_{\mathcal{S}} \subseteq \mathcal{R}(\nu)$ for a c_0 -valued vector measure ν . The coordinate scalar measures $\mu_n = e_n^* \nu$ satisfy $\mu_n(A) \rightarrow 0$ for every measurable set A . Nikodym boundedness and a control-measure construction turn them into a bounded weakly null sequence (g_n) in some $L_1(\lambda)$.

The range containment of all Schreier vectors forces the L_1 -norms of Schreier averages of (g_n) to stay at least $1/2$. This contradicts Szlenk's theorem that $L_1(\lambda)$ has the weak Banach-Saks property.

Proof

Lemma 1. $K_{\mathcal{S}}$ is weakly compact in c_0 .

Proof. Identify subsets of $\mathbb{N}$ with points of $\{0, 1\}^{\mathbb{N}}$. The family $\mathcal{S}$ is closed in $\{0, 1\}^{\mathbb{N}}$. Indeed, if a net from $\mathcal{S}$ converged pointwise to an infinite set M , and $m = \min M$, then eventually the first $m+1$ elements of M would all belong to the approximating finite set, contradicting $|F| \leq \min F = m$. Thus $\mathcal{S}$ is compact.

On bounded subsets of c_0 , the weak topology agrees with coordinatewise convergence. One implication is immediate. Conversely, if a bounded net converges coordinatewise, then it converges against every ℓ_1 functional by dominated convergence. Hence the compact set $K_{\mathcal{S}}$, equipped with coordinatewise convergence, is weakly compact in c_0 . $\square$

Lemma 2 (necessary L_1 domination). *Let $\nu : \Sigma \rightarrow c_0$ be a countably additive vector measure and put $\mu_n = e_n^* \nu$. If $K_{\mathcal{S}} \subseteq \mathcal{R}(\nu)$, then there is a finite positive measure λ and a bounded weakly null sequence $(g_n) \subseteq L_1(\lambda)$ such that, for every finitely supported non-negative scalar sequence (a_n) ,*

$$\sup_{F \in \mathcal{S}} \sum_{n \in F} a_n \leq \left\| \sum_n a_n g_n \right\|_{L_1(\lambda)}.$$

Proof. Since $\nu(A) \in c_0$ for every $A \in \Sigma$, we have

$$\mu_n(A) \rightarrow 0 \quad \text{for every } A \in \Sigma.$$

By the Nikodym boundedness theorem,

$$M := \sup_n |\mu_n|(\Omega) < \infty.$$

Define a finite positive measure

$$\lambda = \sum_{n=1}^{\infty} 2^{-n} \frac{|\mu_n|}{1 + |\mu_n|(\Omega)}.$$

Then each μ_n is absolutely continuous with respect to λ . Let $g_n = d\mu_n/d\lambda$. We have $\|g_n\|_{L_1(\lambda)} = |\mu_n|(\Omega) \leq M$.

For every $A \in \Sigma$,

$$\int_A g_n d\lambda = \mu_n(A) \rightarrow 0.$$

Therefore $\int s g_n d\lambda \rightarrow 0$ for every simple function s . Since simple functions are uniformly dense in $L_{\infty}(\lambda)$ and the g_n 's are L_1 -bounded, it follows that $g_n \rightarrow 0$ weakly in $L_1(\lambda)$.

Now fix $F \in \mathcal{S}$. By the range-containment assumption, choose $A_F \in \Sigma$ such that $\nu(A_F) = 1_F$. Thus

$$\mu_n(A_F) = \begin{cases} 1, & n \in F, \\ 0, & n \notin F. \end{cases}$$

For any finitely supported non-negative scalar sequence (a_n) ,

$$\sum_{n \in F} a_n = \int_{A_F} \sum_n a_n g_n d\lambda \leq \left\| \sum_n a_n g_n \right\|_{L_1(\lambda)}.$$

Taking the supremum over $F \in \mathcal{S}$ proves the domination estimate. $\square$

Proof of the theorem. The weak compactness has already been proved. Suppose, toward a contradiction, that $K_{\mathcal{S}} \subseteq \mathcal{R}(\nu)$ for some c_0 -valued vector measure ν . Let $(g_n) \subseteq L_1(\lambda)$ be the bounded weakly null sequence supplied by the previous lemma.

By Szlenk's weak Banach-Saks theorem for L_1 , there is a subsequence (g_{n_j}) such that

$$\left\| \frac{1}{m} \sum_{j=1}^m g_{n_j} \right\|_{L_1(\lambda)} \longrightarrow 0.$$

We show that this is impossible. Fix $m \geq 2$, set $\ell = \lfloor m/2 \rfloor + 1$, and let

$$F_m = \{n_\ell, n_{\ell+1}, \dots, n_m\}.$$

Then

$$|F_m| = m - \ell + 1 \leq \ell \leq n_\ell = \min F_m,$$

so $F_m \in \mathcal{S}$. Applying the domination lemma to the coefficients

$$a_n = \begin{cases} 1/m, & n \in \{n_1, \dots, n_m\}, \\ 0, & \text{otherwise,} \end{cases}$$

gives

$$\left\| \frac{1}{m} \sum_{j=1}^m g_{n_j} \right\|_{L_1(\lambda)} \geq \sum_{n \in F_m} a_n = \frac{|F_m|}{m} \geq \frac{1}{2}.$$

This contradicts the norm convergence of the Cesaro means to 0. Hence no such vector measure ν exists. $\square$

Consequences for Problem 4.2

The set $K_{\mathcal{S}}$ is a weakly compact subset of c_0 which is not contained in the range of any c_0 -valued vector measure. Thus the Anantharaman-Diestel problem quoted as Problem 4.2 in the source paper has a negative answer.

This also explains why the simpler positive examples are misleading. The set $\{0, e_1, e_2, \dots\}$ can be put in a range using Rademacher functions, and norm-compact subsets can be put in a range by coordinate boxes. The Schreier compactum forces uniformly large finite Schreier averages at arbitrarily far coordinates; this is exactly what the weak Banach-Saks theorem in L_1 rules out for coordinate measures of a hypothetical range.

Novelty and Search Notes

Before promoting this packet, the run indexes were searched for arXiv:1607.07787, “Anantharaman-Diestel”, “weakly compact subset of c_0 ”, “range of a c_0 -valued vector measure”, “Schreier”, and vector-measure range keywords. The only exact local hit was the earlier partial packet for norm-compact sets and $\{0, e_n\}$.

External searches on June 15, 2026 for exact phrases from Problem 4.2 and for Anantharaman-Diestel/vector-measure range variants did not reveal a later published answer. A related 2024 paper of Rodriguez on vector measures with values in $c_0(\kappa)$ was inspected; it records useful coordinate criteria for $c_0(\kappa)$ -valued vector measures, but it does not answer Problem 4.2. The present proof uses only classical tools: Nikodym boundedness, Radon-Nikodym densities with respect to a common finite control measure, and Szlenk's weak Banach-Saks theorem for L_1 .

Human Review Recommendation

The critical line to check is the necessary L_1 -domination lemma. In particular, verify that pointwise null coordinate measures of a c_0 -valued vector measure become a bounded weakly null sequence in a single $L_1(\lambda)$, and that range containment of every Schreier vector gives the stated lower estimate. The final contradiction with Szlenk's theorem is then immediate.

References

- [1] J. Rodriguez, *Open problems in Banach spaces and measure theory*, arXiv:1607.07787.
- [2] J. Diestel and J. J. Uhl, Jr., *Vector Measures*, Mathematical Surveys, No. 15, American Mathematical Society, 1977.
- [3] J. Schreier, *Ein Gegenbeispiel zur Theorie der schwachen Konvergenz*, *Studia Math.* **2** (1930), 58–62.
- [4] W. Szlenk, *Sur les suites faiblement convergentes dans l'espace L* , *Studia Math.* **25** (1965), 337–341.
- [5] J. Rodriguez, *On vector measures with values in $c_0(\kappa)$* , arXiv:2404.05407.